Yue Zhao

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Research Summary

AI systems are deployed faster than their risks can be measured or controlled. My lab builds methods, benchmarks, and open-source tools for **auditing and controlling AI risk** across the deployment stack: *anomaly and out-of-distribution detection in data*, *trust and robustness evaluation of foundation models*, and *audit and control of agent systems across their lifecycle*. The current frontier is turning that agent work from research into **deployable open-source infrastructure**, auditing agents from before deployment to after a run.

1. Agent Layer: Risk Audit and Control

At the agent layer, deployed systems act through tools, APIs, and shared state, making behavior the primary failure surface. This thread covers auditability frameworks, runtime control surfaces, agent-specific failure modes, and post-run attribution and recovery, spanning the full lifecycle from pre-deployment to after a run.

- AI Agent Auditing
- Agent Systems
- Multi-Agent Systems
- Behavior Verification
- AI Agent Security
- Runtime Guardrails
- Policy Enforcement

2. Foundation-Model Layer: Trust and Robustness

Foundation models add a new failure surface between data and agents: behavior under attack, hallucination, and adversarial robustness of retrieval. My lab measures these surfaces through jailbreak detection for vision-language models, causal analysis of hallucination, retrieval-augmented generation attacks, and LLM-as-anomaly-detector benchmarks.

- Foundation-Model Trust
- Trustworthiness Benchmarks
- LLM Hallucination
- Jailbreak Detection
- RAG Security
- Adversarial Robustness
- Multimodal Trust

3. Data Layer: Anomaly and Out-of-Distribution Detection

All audit grounds in distinguishing in-distribution from out-of-distribution behavior. This is my methodological root, started with the PyOD ecosystem and ADBench, and continuing through benchmarks for NLP, automatic OOD detector selection, and modality-specific OOD methods for graphs, multimodal data, and video.

- Anomaly Detection
- Outlier Detection
- Out-of-Distribution Detection
- Distribution Shift
- Multimodal OOD
- Graph OOD
- Benchmark Design

Tools from this group have been cited in a U.S. Senate committee report, a NIST special publication, and the U.S. Department of Defense Generative AI Toolkit. PyOD was selected by the European Space Agency for spacecraft anomaly detection and is the subject of five published books. TrustLLM serves as an official benchmark in all three editions of the Future of Life Institute AI Safety Index.

Professional Experience

University of Southern California
Thomas Lord Department of Computer Science
Assistant Professor (Tenure-Track) — Aug. 2023 - Present

- Foundations Of Robust Trustworthy Intelligent Systems (**FORTIS**) Lab: Link
- USC Machine Learning Center (MaSCle): Link

Fig Work
AI-powered marketplace for early-career talent
Chief Scientific Advisor — Jun. 2026 - Present

PwC Canada
Consulting & Deals
Senior Consultant (Data Scientist) — Aug. 2017 - Jun. 2019
Consultant (Data Scientist) — Feb. 2017 - Jul. 2017
Research Associate (Intern) — May. 2016 - Jan. 2017

Education

Carnegie Mellon University — Pittsburgh, PA
Ph.D. in Information Systems and Management — Sep. 2019 - May. 2023

- **Affiliation**: CMU automated learning systems group (Catalyst) and Data Analytics Techniques Algorithms (DATA) Lab
- **Advisors and Mentors**: CMU: Prof. Leman Akoglu, Prof. Zhihao Jia, and Prof. George Chen. I collaborate with Prof. Jure Leskovec at Stanford, and Prof. Philip S. Yu at UIC.

University of Toronto — Toronto, ON
Master of Science in Computer Science — Sep. 2015 - Dec. 2016

University of Cincinnati — Cincinnati, OH
Bachelor of Science in Computer Engineering — Sep. 2010 - May. 2015
Minor: *Computer Science* and *Mathematics*

Awards, Grants, and Funding

As Principal Investigator (August 2023 onwards)

Foresight Institute AI for Safety & Science Nodes	*Gift*	Apr. 2026
NVIDIA Academic Grant Program	*Gift*	Mar. 2026
Amazon Research Awards, Fall 2025	*Gift*	Mar. 2026
USC CCSL Seed Funding	*Small Grant*	Jan. 2026
Second Prize CCC Award @ IEEE ICDM, BlueSky Track	*Recognition*	Nov. 2025
Best Short Paper Award @ ACM SIGSPATIAL	*Recognition*	Nov. 2025
NSF POSE I	*Funding*	Aug. 2025
Capital One Research Awards	*Grant*	Oct. 2024
Amazon Research Awards, Spring 2024	*Gift*	Aug. 2024
Best Paper Award @ KDD Resource-Efficient Learning Workshop	*Recognition*	Aug. 2024
NSF ATD	*Funding*	Aug. 2024
NSF POSE II	*Funding*	Jun. 2024
Google Cloud Research Innovators	*Recognition*	Mar. 2024
AAAI New Faculty Highlights	*Recognition*	Feb. 2024

Note: Monetary values represent ***my portion*** *of the funding. Total project budgets may be larger.*

Prior to Principal Investigator Role (Before August 2023)

Meta 2022 AI4AI Research Award (student co-PI)	*Recognition*	Oct. 2022
The Norton Labs Graduate Fellowship	*Fellowship*	Mar. 2022
CMU Presidential Fellowship	*Fellowship*	2019
Mitacs-Accelerate Research and Development Funding	*Funding*	2016-2017
University Global Award and Scholarship	*Scholarship*	2010-2015
Mantei/Mae Award & Scholar	*Award*	2012-2015
Engineer of the Month	*Recognition*	Jun. 2014

Note: Monetary values are omitted for awards and recognitions received prior to PI role.

Selected Open-Source Software, Benchmarks, and Systems

Representative open-source research artifacts spanning anomaly detection, trustworthy AI, agent security, and research tooling.

Primary Contributions

- **PyOD** (*Library*; updated Jun. 2026; 9.9K stars) PyOD 3 is the most comprehensive Python library for anomaly detection: 60+ detectors across tabular, time series, graph, text, and image data, with an ADEngine orchestration core and an agentic investigation layer driven through natural language. 42M+ downloads. Links: docs — paper.
- **Anomaly-Detection-Resources** (*Resource Hub*; updated Mar. 2026; 9.3K stars) A curated resource hub for anomaly detection, including papers, benchmarks, tutorials, datasets, tools, and community references.
- **CS-Paper-Checklist** (*Research Toolkit*; updated May. 2026; 1.6K stars) A practical sanity checklist for improving CS paper quality, structure, and submission readiness.
- **PyGOD** (*Library*; updated Nov. 2024; 1.5K stars) A Python library for graph outlier detection, with unified APIs and broad coverage of graph anomaly detection algorithms. Links: docs — paper.
- **ADBench** (*Benchmark*; updated Jan. 2026; 1.0K stars) Official benchmark for anomaly detection with large-scale evaluations across unsupervised, semi-supervised, and supervised settings. Links: paper.
- **combo** (*Library*; updated Jan. 2023; 662 stars) A Python toolbox for machine learning model and score combination across classification, clustering, and anomaly detection tasks. Links: docs — paper.
- **agent-style** (*Developer Toolkit*; updated Jun. 2026; 538 stars) A drop-in rules package that shapes how AI coding and writing agents produce prose at generation time, using 21 curated English style rules (12 from canonical writing authorities; 9 from field observation of LLM output). Works with Claude Code, Codex, Copilot, Cursor, Aider, and other AGENTS.md-compliant tools. Links: docs — demo.
- **SUOD** (*Library / System*; updated Mar. 2025; 394 stars) An acceleration framework for large-scale unsupervised heterogeneous outlier detection with efficient training and inference. Links: docs — paper.
- **Aegis** (*Security / Compliance Platform*; updated Jun. 2026; 360 stars) The open-source firewall for AI agents, with pre-execution policy checks, human-in-the-loop approvals, and tamper-evident audit trails for safer and more accountable deployment. Links: docs — paper — demo.
- **agent-audit** (*Security Toolkit*; updated Jun. 2026; 192 stars) An open-source tool for auditing AI agent code and configurations, with OWASP Agentic Top 10 coverage, taint analysis, and MCP configuration auditing. Links: docs — paper — demo.
- **anywhere-agents** (*Developer Toolkit*; updated Jun. 2026; 186 stars) One config to rule all your AI agents: portable across every project and session, effective through curated writing, routing, and skills, and safer via a PreToolUse guard that blocks destructive Git and GitHub commands. Supports Claude Code and Codex today, with plans to grow. Links: docs.
- **AD-AGENT** (*Platform*; updated Apr. 2026; 100 stars) An LLM-driven multi-agent anomaly detection platform for end-to-end workflows across tabular, graph, and time-series settings. Links: docs — paper.
- **AD-LLM** (*Benchmark*; updated Oct. 2025; 45 stars) A benchmark for using large language models in anomaly detection, covering zero-shot detection, data augmentation, and model selection. Links: paper.
- **NLP-ADBench** (*Benchmark*; updated Oct. 2025; 21 stars) A benchmark for anomaly detection in NLP, including NLPAD datasets and standardized evaluation across end-to-end and embedding-based methods. Links: paper.
- **auditable** (*Agent Auditing*; updated Jun. 2026; 12 stars) An open-source system of record for AI-agent decisions: it captures what each decision relied on, replays it against live state and rolls back the committed

action when it no longer holds, and ranks the steps of a finished run by structural blast share, on one typed graph across an agent's plan, run, and review. Links: docs — paper — demo.

- **awesome-auditable-ai** (*Resource Hub*; updated Jun. 2026; 8 stars) A curated list of papers, tools, datasets, benchmarks, and standards for building, evaluating, and auditing reliable AI agents, organized around a reader-first reliability map.

Selected Collaborations

- **TODS** (*System / AutoML*; updated Sep. 2023; 1.7K stars) An automated time-series outlier detection system with end-to-end pipeline support for preprocessing, feature extraction, model search, and detection. Links: docs — paper.
- **TDC** (*Benchmark / Platform*; updated Jul. 2025; 1.3K stars) Therapeutics Data Commons, a platform for AI-ready therapeutic datasets, benchmarks, and evaluation workflows across drug discovery tasks. Links: docs — paper.
- **TrustLLM** (*Benchmark / Toolkit*; updated Jun. 2025; 628 stars) A trustworthiness benchmark and toolkit for LLMs; collaboration included follow-on NSF-supported development with the core team. Links: docs — paper.
- **TrustEval-toolkit** (*Benchmark / Toolkit*; updated Aug. 2025; 131 stars) A toolkit and benchmark for evaluating trustworthiness of generative foundation models across LLM, VLM, and text-to-image settings. Links: docs — paper.
- **LangSkills** (*Toolkit / Knowledge Base*; updated Mar. 2026; 118 stars) An evidence-backed skill library and pipeline for AI agents, with offline searchable bundles distilled from papers and technical sources.

Publications

G Scholar

ResearchG

Preprints & Under Submission Note: †Equal contribution and ♠Corresponding author for multi-first/corresponding author papers.

114. Yue Zhao
GRADE: Graph Representation of LLM Agent Dependency and Execution
arXiv preprint arXiv:2606.22741
113. Wenjie Wang, Yue Huang, Zhengqing Yuan, Han Bao, Shiyi Du, Yuchen Ma, Yue Zhao, Yanfang Ye, Xiangliang Zhang
SpecAlign: Efficient Specification-Grounded Alignment of Large Language Models via Synthetic Data
arXiv preprint arXiv:2606.16276
112. Yi Nian, Tiankai Yang, Yudi Zhang, Qi Pan, Zelong Xu, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, Yue Zhao
DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment
arXiv preprint arXiv:2606.07678
111. Yingzi Ma, Zhengyue Zhao, Xiaogeng Liu, Minhui Xue, Yue Zhao, Chaowei Xiao
MaskForge: Structure-Aware Adaptive Attacks for Jailbreaking Diffusion Large Language Models
arXiv preprint arXiv:2606.04027
110. Yuehan Qin, Li Li, Linxin Song, Wei Yang, Jiate Li, Yuqing Yang, Yue Zhao
Memory Retrieval for Changing Preferences
arXiv preprint arXiv:2606.02976
109. Ojas Nimase, Jiate Li, Yue Zhao, Yushun Dong
Can Subgraph Explanations Be Weaponized to Steal Graph Neural Networks?
arXiv preprint arXiv:2605.30470
108. Ojas Nimase, Zhe Chen, Gengpei Qi, Yue Zhao, Xiyang Hu
GEO-Bench: Benchmarking Ranking Manipulation in Generative Engine Optimization
arXiv preprint arXiv:2605.29107
107. Haolin Chen, Deon Metelski, Leon Qi, Tao Xia, Joonyul Lee, Steve Brown, Kevin Riley, Frank Wang, T. Y. Alvin Liu, Hank Capps MD, Zeyu Tang, Xiangchen Song, Lingjing Kong, Fan Feng, Tianyi Zeng, Zhiwei Liu, Zixian Ma, Hang Jiang, Fangli Geng, Yuan Yuan, Chenyu You, Qingsong Wen, Hua Wei, Yanjie Fu, Yue Zhao, Carl Yang, Biwei Huang, Kun Zhang, Caiming Xiong, Sanmi Koyejo, Eric P. Xing, Philip S. Yu, Weiran Yao
CHI-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?
arXiv preprint arXiv:2605.16679

106. Shawn Li, Chenxiao Yu, Han Wang, Wei Yang, Ryan Rossi, Franck Dernoncourt, Xiyang Hu, Philip Yu, Chaowei Xiao, Huan Zhang, Yue Zhao
FORTIS: Benchmarking Over-Privilege in Agent Skills
arXiv preprint arXiv:2605.09163

105. Yicheng Gao, Xiaolin Zhou, Yahan Li, Yue Zhao, Ruishan Liu
MedExAgent: Training LLM Agents to Ask, Examine, and Diagnose in Noisy Clinical Environments
arXiv preprint arXiv:2605.07058

104. Shawn Li, You Qin, Jiate Li, Charith Peris, Lisa Bauer, Roger Zimmermann, Yue Zhao
Geometry over Density: Few-Shot Cross-Domain OOD Detection
arXiv preprint arXiv:2605.03410

103. Tiankai Yang, Yi Nian, Xinyuan Li, Ruiyao Xu, Kaize Ding, Yue Zhao
Cat-DPO: Category-Adaptive Safety Alignment
arXiv preprint arXiv:2604.17299

102. Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan Rossi, Kaize Ding, Yue Zhao
No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents
arXiv preprint arXiv:2604.01350

101. Haiyue Zhang, Yi Nian, Yue Zhao
Agent Audit: A Security Analysis System for LLM Agent Applications
arXiv preprint arXiv:2603.22853

100. Shawn Li, Yue Zhao
The Autonomy Tax: Defense Training Breaks LLM Agents
Under submission
arXiv preprint arXiv:2603.19423

99. Yi Nian, Haosen Cao, Shenzhe Zhu, Henry Peng Zou, Qingqing Luan, Yue Zhao
When Only the Final Text Survives: Implicit Execution Tracing for Multi-Agent Attribution
Under submission
arXiv preprint arXiv:2603.17445

98. Aojie Yuan, Zhiyuan Su, Yue Zhao
AEGIS: No Tool Call Left Unchecked – A Pre-Execution Firewall and Audit Layer for AI Agents
Under submission
arXiv preprint arXiv:2603.12621

97. Rujie Ye, Jiayi Zhang, Zhuoxin Liu, Zihao Zhu, Siyuan Yang, Li Li, Tianfu Fu, Franck Dernoncourt, Yue Zhao, Jiacheng Zhu, Ryan Rossi, Wenhao Chai, Zhengzhong Tu
Agent Banana: High-Fidelity Image Editing with Agentic Thinking and Tooling
Under submission
arXiv preprint arXiv:2602.09084

96. Xiaolin Zhou, Zheng Luo, Yicheng Gao, Qixuan Chen, Xiyang Hu, Yue Zhao, Ruishan Liu
Fairness or Fluency? An Investigation into Language Bias of Pairwise LLM-as-a-Judge
Under submission
arXiv preprint arXiv:2601.13649

95. Chenxiao Yu, Bowen Yi, Farzan Karimi-Malekabadi, Suhaib Abdurahman,
Jinyi Ye, Shrikanth Narayanan, Yue Zhao, Morteza Dehghani
Tracing Moral Foundations in Large Language Models
Under submission
arXiv preprint arXiv:2601.05437

94. Kay Liu, Yuwei Han, Haoyan Xu, Henry Peng Zou, Yue Zhao, Philip S. Yu
TAGFN: A Text-Attributed Graph Dataset for Fake News Detection in the Age of LLMs
Under submission
arXiv preprint arXiv:2511.21624

93. Haoyan Xu, Ruizhi Qian, Zhengtao Yao, Ziyi Liu, Li Li, Yuqi Li, Yanshu Li, Wenqing Zheng, Daniele Rosa, Daniel Barcklow, Senthil Kumar, Jieyu Zhao, Yue Zhao
LLM-Powered Text-Attributed Graph Anomaly Detection via Retrieval-Augmented Reasoning
Under submission
arXiv preprint arXiv:2511.17584

92. Haoyan Xu, Ruizhi Qian, Jiate Li, Yushun Dong, Minghao Lin, Hanson Yan, Zhengtao Yao, Qinghua Liu, Junhao Dong, Ruopeng Huang, Yue Zhao♠, Mengyuan Li♠
A Systematic Study of Model Extraction Attacks on Graph Foundation Models
Under submission
arXiv preprint arXiv:2511.11912

91. Yuexing Hao, Yue Huang, Haoran Zhang, Chenyang Zhao, Zhenwen Liang, Paul Pu Liang, Yue Zhao, Lichao Sun, Saleh Kalantari, Xiangliang Zhang, Marzyeh Ghassemi
The Role of Computing Resources in Publishing Foundation Model Research
Under submission
arXiv preprint arXiv:2510.13621

90. Wang Wei, Tiankai Yang, Hongjie Chen, Yue Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry
Learning to Route LLMs from Bandit Feedback: One Policy, Many Trade-offs
Under submission
arXiv preprint arXiv:2510.07429

89. Langzhou He, Junyou Zhu, Fangxin Wang, Junhua Liu, Haoyan Xu, Yue Zhao, Philip S. Yu, Qitian Wu
Can Molecular Foundation Models Know What They Don't Know? A Simple Remedy with Preference Optimization
Under submission
arXiv preprint arXiv:2509.25509

88. Yuehan Qin, Li Li, Defu Cao, Tiankai Yang, Yue Zhao
M3OOD: Automatic Selection of Multimodal OOD Detectors
Under submission
arXiv preprint arXiv:2508.11936

87. Haoyan Xu, Zhengtao Yao, Xuzhi Zhang, Ziyi Wang, Langzhou He, Yushun Dong, Philip S. Yu, Mengyuan Li, Yue Zhao
GLIP-OOD: Zero-Shot Graph OOD Detection with Foundation Model
Under submission
arXiv preprint arXiv:2504.21186

86. Haoyan Xu, Zhengtao Yao, Ziyi Wang, Zhan Cheng, Xiyang Hu, Mengyuan Li, Yue Zhao
Graph Synthetic Out-of-Distribution Exposure with Large Language Models
Under submission
arXiv preprint arXiv:2504.21198

85. Kaixiang Zhao, Lincan Li, Kaize Ding, Neil Zhenqiang Gong, Yue Zhao, Yushun Dong
A Survey of Model Extraction Attacks and Defenses in Distributed Computing Environments
Under submission
arXiv preprint arXiv:2502.16065

84. Shixuan Li, Wei Yang, Peiyu Zhang, Xiongye Xiao, Defu Cao, Yuehan Qin, Xiaole Zhang, Yue Zhao, Paul Bogdan
ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models
Under submission
arXiv preprint arXiv:2502.11059

83. Lincan Li, Jiaqi Li, Catherine Chen♠, Fred Gui♠, other collaborators, Yue Zhao♠, Yushun Dong♠
Political-LLM: Large Language Models in Political Science
Under submission
arXiv preprint arXiv:2412.06864

82. Chenxiao Yu, Jinyi Ye, Yuangang Li, Zhaotian Weng, Zheng Li, Emilio Ferrara, Xiyang Hu♠, Yue Zhao♠
A Large-Scale Simulation on Large Language Models for Decision-Making in Political Science
Under submission
arXiv preprint arXiv:2412.15291

81. Junda Wu, Hanjia Lyu, Yu Xia, Zhehao Zhang, Joe Barrow, Ishita Kumar, Mehnoosh Mirtahebi, Hongjie Chen, Ryan A. Rossi, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, Jiuxiang Gu, Nesreen K. Ahmed, Yu Wang, Xiang Chen, Hanieh Deilamsalehy, Namyong Park, Sungchul Kim, Huanrui Yang, Subrata Mitra, Zhengmian Hu, Nedim Lipka, Yue Zhao, Jiebo Luo, Julian McAuley
Personalized Multimodal Large Language Models: A Survey
Under submission
arXiv preprint arXiv:2412.02142

Peer-reviewed Journal Papers

80. Yuehan Qin, Shawn Li, Yi Nian, Xinyan Velocity Yu, Yue Zhao♠, Xuezhe Ma♠
Don't Let It Hallucinate: Premise Verification via Retrieval-Augmented Logical Reasoning
Transactions on Machine Learning Research (TMLR), 2026
79. Haoyan Xu, Kay Liu, Zhengtao Yao, Philip S. Yu, Kaize Ding♠, Yue Zhao♠
LEGO-Learn: Label-Efficient Graph Open-Set Learning
*Transactions on Machine Learning Research (**TMLR**)*, 2025
78. Hao Dong, Gaetan Frusque, Yue Zhao, Eleni Chatzi, Olga Fink
NNG-Mix: Improving Semi-supervised Anomaly Detection with Pseudo-anomaly Generation
*IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)*, 2024
77. Ling Yang†, Zhilong Zhang†, Yang Song, Shenda Hong, Runsheng Xu, Yue Zhao, Wentao Zhang, Bin Cui, Ming-Hsuan Yang
Diffusion Models: A Comprehensive Survey of Methods and Applications
*ACM Computing Surveys (**CSUR**)*, 2023
76. Yue Zhao†, Martin Q. Ma†, Xiaorong Zhang, Leman Akoglu
The Need for Unsupervised Outlier Model Selection: A Review and Evaluation of Internal Evaluation Strategies
*ACM SIGKDD Explorations Newsletter (**SIGKDD Explor.**)*, 2023
75. Kexin Huang†, Tianfan Fu†, Wenhao Gao†, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Artificial Intelligence Foundation for Therapeutic Science
*Nature Chemical Biology (**NCHEMB**)*, 2022
74. Yue Zhao†, Zheng Li†, Xiyang Hu, Nicola Botta, Cezar Ionescu, George H. Chen
ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions
*IEEE Transactions on Knowledge and Data Engineering (**TKDE**)*, 2022.
73. Yue Zhao, Zain Nasrullah, Zheng Li
PyOD: A Python Toolbox for Scalable Outlier Detection
Journal of Machine Learning Research (***JMLR***), 2019.

Conference & Workshop Papers

72. Yiming Tang, Yi Fan, Chenxiao Yu, Tiankai Yang, Yue Zhao, Xiyang Hu
StealthRank: LLM Ranking Manipulation via Stealthy Prompt Optimization
***ICML Workshop** on Trustworthy AI for Good (**AI4GOOD**)*, 2026.
71. Han Bao, Yue Huang, Yanbo Wang, Jiayi Ye, Xiangqi Wang, Xiuying Chen, Yue Zhao, Tianyi Zhou, Mohamed Elhoseiny, Xiangliang Zhang
AutoDavis: Automatic and Dynamic Evaluation Protocol of Large Vision-Language Models on Visual Question-Answering
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Datasets and Benchmarks Track)*, 2026.
70. Zhisheng Qi, Utkarsh Sahu, Li Ma, Haoyu Han, Ryan Rossi, Franck Dernoncourt, Mahantesh Halappanavar, Nesreen Ahmed, Yushun Dong, Yue Zhao, Yu Zhang, Yu Wang
Benchmarking Knowledge-Extraction Attack and Defense on Retrieval-Augmented Generation
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Datasets and Benchmarks Track)*, **Oral**, 2026.
69. Jiate Li, Defu Cao, Li Li, Wei Yang, Yuehan Qin, Chenxiao Yu, Tiannuo Yang, Ryan A. Rossi, Yan Liu, Xiyang Hu, Yue Zhao
"Someone Hid It": Query-Agnostic Black-Box Attacks on LLM-Based Retrieval
*International Conference on Machine Learning (**ICML**)*, 2026.
68. Shawn Li, Chenxiao Yu, Zhiyu Ni, Hao Li, Charith Peris, Chaowei Xiao, Yue Zhao
Defenses Against Prompt Attacks Learn Surface Heuristics
*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026.
67. Ruiyao Xu, Mihir Parmar, Tiankai Yang, Zhengyu Hu, Yue Zhao, Kaize Ding
CoAct: Co-Active LLM Preference Learning with Human-AI Synergy
*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, **Oral**, 2026.

66. Jinbo Liu, Defu Cao, Yifei Wei, Tianyao Su, Yuan Liang, Yushun Dong, Yan Liu, Yue Zhao, Xiyang Hu
Topology Matters: Measuring Memory Leakage in Multi-Agent LLMs
Findings *of the Association for Computational Linguistics:* ***ACL***, 2026.

65. Yi Nian, Aojie Yuan, Haiyue Zhang, Jiate Li, Yue Zhao
Auditable Agents
ACL Workshop *on Towards Knowledgeable Foundation Models (**KnowFM**)*, 2026.
arXiv preprint arXiv:2604.05485

64. Yixuan Du, Chenxiao Yu, Haoyan Xu, Ziyi Wang, Yue Zhao, Xiyang Hu
Multimodal Generative Engine Optimization: Exploiting Cross-Modal Knowledge Grounding in VLM-Based Ranking
ACL Workshop *on Towards Knowledgeable Foundation Models (**KnowFM**)*, 2026.
arXiv preprint arXiv:2601.12263

63. Shawn Li, Ryan Rossi, Sungchul Kim, Sunav Choudhary, Franck Dernoncourt, Puneet Mathur, Zhengzhong Tu, Yue Zhao
Charts Are Not Images: On the Challenges of Scientific Chart Editing
*International Conference on Learning Representations (**ICLR**)*, 2026

62. Weidi Luo, Tianyu Lu, Qiming Zhang, Xiaogeng Liu, Bin Hu, Yue Zhao, Jieyu Zhao, Song Gao, Patrick McDaniel, Zhen Xiang, Chaowei Xiao
Doxing via the Lens: Revealing Privacy Leakage in Image Geolocation for Agentic Multi-Modal Large Reasoning Model
*International Conference on Learning Representations (**ICLR**)*, 2026

61. Yue Huang, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Yujun Zhou, Yanbo Wang, Jiayi Ye, Jiawen Shi, Qihui Zhang, Yuan Li, Han Bao, Zhaoyi Liu, Tianrui Guan, Dongping Chen, Ruoxi Chen, other authors, Yue Zhao, other authors, Xiangliang Zhang
On the Trustworthiness of Generative Foundation Models: Guideline, Assessment, and Perspective
*International Conference on Learning Representations (**ICLR**)*, 2026
`https://trustgen.github.io/`

60. Chengxuan Qian, Shuo Xing, Shawn Li, Yue Zhao, Zhengzhong Tu
DecAlign: Hierarchical Cross-Modal Alignment for Decoupled Multimodal Representation Learning
*International Conference on Learning Representations (**ICLR**)*, 2026

59. Xiongxiao Xu, Haoran Wang, Yueqing Liang, Philip S. Yu, Yue Zhao, Kai Shu
Can Multimodal LLMs Perform Time Series Anomaly Detection?
*The Web Conference (**WWW**)*, 2026

58. Bo Ni, Yu Wang, Leyao Wang, Branislav Kveton, Franck Dernoncourt, Yu Xia, Hongjie Chen, Reuben Luera, Samyadeep Basu, Subhojyoti Mukherjee, Puneet Mathur, Nesreen K. Ahmed, Junda Wu, Shawn Li, Huixin Zhang, Ruiyi Zhang, Tong Yu, Sungchul Kim, Jiuxiang Gu, Zhengzhong Tu, Alexa Siu, Zichao Wang, Seunghyun Yoon, Nedim Lipka, Namyong Park, Zihao Lin, Trung Bui, Yue Zhao, Tyler Derr, Ryan A. Rossi
A Survey on LLM-based Conversational User Simulation
*Conference of the European Chapter of the Association for Computational Linguistics (**EACL**)*, 2026

57. Yuangang Li, Yiqing Shen, Yi Nian, Jiechao Gao, Ziyi Wang, Chenxiao Yu, Shawn Li, Jie Wang, Xiyang Hu, Yue Zhao
Mitigating Hallucinations in Large Language Models via Causal Reasoning
*Fortieth AAAI Conference on Artificial Intelligence (**AAAI**)*, 2026

56. Ojas Nimase, Yue Zhao, Yushun Dong
Navigating Between Explainability and Extractability in Machine Learning as a Service
*IEEE International Conference on Data Mining (**ICDM**) BlueSky Track*, Second Prize CCC Award, 2025.

55. Tiankai Yang, Junjun Liu, Michael Siu, Jiahang Wang, Zhuangzhuang Qian, Chanjuan Song, Cheng Cheng, Xiyang Hu, Yue Zhao
AD-AGENT: A Multi-agent Framework for End-to-end Anomaly Detection
Findings *of the Association for Computational Linguistics:* ***IJCNLP-AACL***, 2025.

54. Ruosi Shao, Md Shamim Seraj, Kangyi Zhao, Yingtao Luo, Lincan Li, Bolin Shen, Averi Bates, Yue Zhao, Chongle Pan, Lisa Hightow-Weidman, Shayok Chakraborty, Yushun Dong
LLM-Empowered Patient-Provider Communication: A Data-Centric Survey From a Clinical Perspective
Findings *of the Association for Computational Linguistics:* ***IJCNLP-AACL***, 2025.

53. Li Li, Peilin Cai, Ryan A. Rossi, Franck Dernoncourt, Branislav Kveton, Junda Wu, Tong Yu, Lixin Song, Tiankai Yang, Yuehan Qin, Nesreen K. Ahmed, Samyadeep Basu, Subhojyoti Mukherjee, Ruiyi Zhang, Yuxiao Zhou, Zichao Wang, Yue Huang, Yu Wang, Xiangliang Zhang, Philip S. Yu, Xiyang Hu, Yue Zhao
A Personalized Conversational Benchmark: Towards Simulating Personalized Conversations
NeurIPS Workshop *on Multi-Turn Interactions in Large Language Models (**MTI-LLM**)*, **Spotlight**, 2025.
arXiv preprint arXiv:2505.14106

52. Zixiang Xu, Yanbo Wang, Yue Huang, Jiayi Ye, Haomin Zhuang, Zirui Song, Lang Gao, Chenxi Wang, Zhaorun Chen, Yujun Zhou, Sixian Li, Wang Pan, Yue Zhao, Jieyu Zhao, Xiangliang Zhang, Xiuying Chen
SocialMaze: A Benchmark for Evaluating Social Reasoning in Large Language Models
NeurIPS Workshop *on Socially Responsible and Trustworthy Foundation Models (**ResponsibleFM**)*
arXiv preprint arXiv:2505.23713

51. Yanbo Wang, Zixiang Xu, Yue Huang, Xiangqi Wang, Zirui Song, Lang Gao, Chenxi Wang, Xiangru Tang, Yue Zhao, Arman Cohan, Xiangliang Zhang, Xiuying Chen
DyFlow: Dynamic Workflow Framework for Agentic Reasoning
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2025

50. Shawn Li, Jiashu Qu, Yuxiao Zhou, Yuehan Qin, Tiankai Yang, Yue Zhao
Treble Counterfactual VLMs: A Causal Approach to Hallucination
Findings *of the Association for Computational Linguistics: **EMNLP***, 2025.

49. Yuangang Li, Jiaqi Li, Zhuo Xiao, Tiankai Yang, Yi Nian, Xiyang Hu, Yue Zhao
NLP-ADBench: NLP Anomaly Detection Benchmark
Findings *of the Association for Computational Linguistics: **EMNLP***, 2025.

48. Lincan Li, Eren Erman Ozguven, Yue Zhao, Guang Wang, Yiqun Xie, Yushun Dong
TyphoFormer: Language-Augmented Transformer for Accurate Typhoon Track Forecasting
*ACM International Conference on Advances in Geographic Information Systems (**SIGSPATIAL**)*, Best Short Paper Award, 2025.

47. Bolin Shen, Eren Erman Ozguven, Yue Zhao, Guang Wang, Yiqun Xie, Yushun Dong
Learning from the Storm: A Multivariate Machine Learning Approach to Predicting Hurricane-Induced Economic Losses
*ACM SIGSPATIAL International Workshop on Spatial Intelligence for Smart and Connected Communities (**SpatialConnect**)*, 2025

46. Yi Nian[†], Shenzhe Zhu[†], Yuehan Qin, Shawn Li, Ziyi Wang, Chaowei Xiao, Yue Zhao
JailDAM: Jailbreak Detection with Adaptive Memory for Vision-Language Model
*Conference on Language Modeling (**COLM**)*, 2025.

45. Shawn Li, Peilin Cai, Yuxiao Zhou, Zhiyu Ni, Renjie Liang, You Qin, Yi Nian, Zhengzhong Tu, Xiyang Hu, Yue Zhao
Secure On-Device Video OOD Detection Without Backpropagation
*International Conference on Computer Vision (**ICCV**)*, 2025.

44. Zerui Xu, Fang Wu, Yue Zhao
Retrieval-Reasoning Large Language Model-based Synthetic Clinical Trial Generation
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (***KDD Workshop*** *on AI Agent for Information Retrieval*), 2025.
ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (***ACM BCB***, 2025.

43. Haoyan Xu[†], Zhengtao Yao[†], Yushun Dong, Ziyi Wang, Ryan A. Rossi, Mengyuan Li, Yue Zhao
Few-Shot Graph Out-of-Distribution Detection with LLMs
*European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML PKDD**)*, 2025.

42. Tiankai Yang[†], Yi Nian[†], Shawn Li, Ruiyao Xu, Yuangang Li, Jiaqi Li, Xiyang Hu, Ryan Rossi, Kaize Ding, Xia Hu, Yue Zhao
AD-LLM: Benchmarking Large Language Models for Anomaly Detection
*Findings of the Association for Computational Linguistics (**ACL** Findings)*, 2025.

41. Yu Xia, Subhojyoti Mukherjee, Zhouhang Xie, Junda Wu, Xintong Li, Ryan Aponte, Hanjia Lyu, Joe Barrow, Hongjie Chen, Franck Dernoncourt, Branislav Kveton, Tong Yu, Ruiyi Zhang, Jiuxiang

Gu, Nesreen K. Ahmed, Yu Wang, Xiang Chen, Hanieh Deilamsalehy, Sungchul Kim, Zhengmian Hu, Yue Zhao, Nedim Lipka, Seunghyun Yoon, Ting-Hao Kenneth Huang, Zichao Wang, Puneet Mathur, Soumyabrata Pal, Koyel Mukherjee, Zhehao Zhang, Namyong Park, Thien Huu Nguyen, Jiebo Luo, Ryan A. Rossi, Julian McAuley
From Selection to Generation: A Survey of LLM-based Active Learning
*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2025.

40. Kaixiang Zhao, Lincan Li, Kaize Ding, Neil Zhenqiang Gong, Yue Zhao, Yushun Dong
A Survey on Model Extraction Attacks and Defenses for Large Language Models
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Lecture-Style Tutorial Track)*, 2025.

39. Shawn Li, Huixian Gong, Hao Dong, Tiankai Yang, Zhengzhong Tu, Yue Zhao
DPU: Dynamic Prototype Updating for Multimodal Out-of-Distribution Detection
*Conference on Computer Vision and Pattern Recognition (**CVPR**)*, **Highlight**, 2025

38. Hanhui Wang, Yihua Zhang, Ruizheng Bai, Yue Zhao, Sijia Liu, Zhengzhong Tu
Edit Away and My Face Will Not Stay: Personal Biometric Defense against Malicious Generative Editing
*Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025

37. Yanbo Wang, Jiayi Ye, Siyuan Wu, Chujie Gao, Yue Huang, Xiuying Chen, Yue Zhao, Xiangliang Zhang
TRUSTEVAL: A Dynamic Evaluation Toolkit on Trustworthiness of Generative Foundation Models
*Annual Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL** Demo Track)*, 2025.

36. Yuehan Qin[†], Yichi Zhang[†], Yi Nian[†], Xueying Ding, Yue Zhao
MetaOOD: Meta-learning for Automatic Out-of-Distribution Detection Model Selection
*International Conference on Learning Representations (**ICLR**)*, 2025
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (***KDD Workshop** on Resource-Efficient Learning for Knowledge Discovery*), Best Paper Award, 2024.

35. Sihan Chen, Zhuangzhuang Qian, Michael Siu, Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Tiankai Yang, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, Yue Zhao
PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection
*International World Wide Web Conference (**The Web Conference** Demo Track)*, 2025

34. Sizhe Liu, Yizhou Lu, Siyu Chen, Xiyang Hu, Jieyu Zhao, Yingzhou Lu, Yue Zhao
DrugAgent: Automating AI-aided Drug Discovery Programming through LLM Multi-Agent Collaboration
***AAAI Workshop** on Foundation Models for Biological Discoveries (**FMs4Bio**)*, 2025.

33. Hao Dong, Yue Zhao, Eleni Chatzi, Olga Fink
MultiOOD: Scaling Out-of-Distribution Detection for Multiple Modalities
*Advances in Neural Information Processing Systems (**NeurIPS**)*, **Spotlight**, 2024

32. Xueying Ding, Yue Zhao, Leman Akoglu
Fast Unsupervised Deep Outlier Model Selection with Hypernetworks
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024*

31. Lichao Sun, Yue Huang, Haoran Wang, Siyuan Wu, Qihui Zhang, Chujie Gao, Yixin Huang, Wenhan Lyu, Yixuan Zhang, Xiner Li, Zhengliang Liu, Yixin Liu, Yijue Wang, Zhikun Zhang, 50+ collaborative authors, Yue Zhao
TrustLLM: Trustworthiness in Large Language Models
*International Conference on Machine Learning (**ICML**)*, 2024

30. Yue Zhao, Leman Akoglu
Hyperparameter Optimization for Unsupervised Outlier Detection
*International Conference on Automated Machine Learning (**AutoML**)*, 2024

29. Yue Zhao
Towards Reproducible, Automated, and Scalable Anomaly Detection
*AAAI Conference on Artificial Intelligence (**AAAI**), New Faculty Highlights*, 2024

28. Minqi Jiang[†], Chaochuan Hou[†], Ao Zheng[†], Songqiao Han, Hailiang Huang[♠], Qingsong Wen, Xiyang Hu[♠], Yue Zhao[♠]
ADGym: Design Choices for Deep Anomaly Detection.
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2023
([♠]Corresponding author)

27. Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu
DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection
*European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML/PKDD**)*, 2023

26. Peng Xu, Lin Zhang, Xuanzhou Liu, Jiaqi Sun, Yue Zhao, Haiqin Yang, Bei Yu
Do Not Train It: A Linear Neural Architecture Search of Graph Neural Networks
*International Conference on Machine Learning (**ICML**)*, 2023

25. Yue Zhao, Guoqing Zheng, Subhabrata Mukherjee, Robert McCann, Ahmed Awadallah
ADMoE: Anomaly Detection with Mixture-of-Experts from Noisy Labels
*Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI**)*, 2023

24. Yue Zhao, George H. Chen, Zhihao Jia
TOD: GPU-accelerated Outlier Detection via Tensor Operations
*International Conference on Very Large Data Bases (**VLDB**)*, 2023

23. Songqiao Han†, Xiyang Hu†, Hailiang Huang†, Minqi Jiang†, Yue Zhao†♠
ADBench: Anomaly Detection Benchmark
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022
(†Equal contribution & ♠Corresponding author)

22. Yue Zhao†, Kay Liu†, Yingtong Dou†, et al.
Benchmarking Node Outlier Detection on Graphs
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022.

21. Yue Zhao, Xiaorong Zhang, Leman Akoglu
ELECT: Toward Unsupervised Outlier Model Selection
*IEEE International Conference on Data Mining (**ICDM**)*, 2022.

20. Zhiming Xu, Xiao Huang, Yue Zhao, Yushun Dong, Jundong Li
Contrastive Attributed Network Anomaly Detection with Data Augmentation
*Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)*, 2022.

19. Yue Zhao, Ryan A. Rossi, Leman Akoglu
Automatic Unsupervised Outlier Model Selection
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021.

18. Kwei-Herng Lai, Daochen Zha, Junjie Xu, Yue Zhao, Guanchu Wang, Xia Hu
Revisiting Time Series Outlier Detection: Definitions and Benchmarks
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

17. Kexin Huang†, Tianfan Fu†, Wenhao Gao†, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

16. Yue Zhao†, Xiyang Hu†, Cheng Cheng, Cong Wang, Changlin Wan, Wen Wang, Jianing Yang, Haoping Bai, Zheng Li, Cao Xiao, Yunlong Wang, Zhi Qiao, Jimeng Sun, Leman Akoglu
SUOD: Accelerating Large-scale Unsupervised Heterogeneous Outlier Detection
*Conference on Machine and Learning Systems (**MLSys**)*, 2021.

15. Kwei-Herng Lai†, Daochen Zha†, Guanchu Wang, Junjie Xu, Yue Zhao, Devesh Kumar, Yile Chen, Purav Zumkhawaka, Minyang Wan, Diego Martinez and Xia Ben Hu
TODS: An Automated Time Series Outlier Detection System (Demo paper)
*Thirty-Fifth AAAI Conference on Artificial Intelligence (**AAAI**)*, 2021.

14. Meng-Chieh Lee, Yue Zhao, Aluna Wang, Pierre Jinghong Liang, Leman Akoglu, Vincent S. Tseng, Christos Faloutsos
AutoAudit: Mining Accounting and Time-Evolving Graphs
*IEEE International Conference on Big Data (**Big Data**)*, 2020

13. Changlin Wan, Dongya Jia, Yue Zhao, Wennan Chang, Sha Cao, Xiao Wang, and Chi Zhang
A Data Denoising Approach to Optimize Functional Clustering of Single Cell RNA-sequencing Data
*IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**)*, 2020

12. Yue Zhao, Xueying Ding, Jianing Yang, Haoping Bai.
SUOD: Toward Scalable Unsupervised Outlier Detection
***Workshops** at the Thirty-Fourth **AAAI** Conference on Artificial Intelligence*, 2020.
Extended version published in *MLSys 2021*.

11. Zheng Li, Yue Zhao, Nicola Botta, Cezar Ionescu, Xiyang Hu
COPOD: Copula-Based Outlier Detection
*IEEE International Conference on Data Mining (**ICDM**)*, 2020.

10. Zheng Li, Yue Zhao, Jialin Fu
SYNC: A Copula based Framework for Generating Synthetic Data from Aggregated Sources
*IEEE International Conference on Data Mining Workshops (**ICDMW**)*, 2020.

9. Yiqun Mei, Yue Zhao, Wei Liang
DSR: An Accurate Single Image Super Resolution Approach for Various Degradations
*IEEE International Conference on Multimedia and Expo (**ICME**)*, 2020, London, UK.

8. Yue Zhao, Xuejian Wang†, Cheng Cheng†, Xueying Ding†
Combining Machine Learning Models and Scores using combo Library (Demo paper)
*Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI**)*, 2020.

7. Colin Wan, Zheng Li, Alicia Guo, Yue Zhao
SynC: A Unified Framework for Generating Synthetic Population with Gaussian Copula
***Workshops** at the Thirty-Fourth **AAAI** Conference on Artificial Intelligence*, 2020.
Extended version published in *ICDMW 2020*.

6. Zain Nasrullah, Yue Zhao
Music Artist Classification with Convolutional Recurrent Neural Networks
IEEE International Joint Conference on Neural Networks (***IJCNN***), 2019, Hungary.

5. Yue Zhao, Zain Nasrullah, Maciej K. Hryniewicki, Zheng Li
LSCP: Locally Selective Combination in Parallel Outlier Ensembles
SIAM International Conference on Data Mining (***SDM***), 2019, Calgary, Canada.

4. Yue Zhao, Maciej K. Hryniewicki
DCSO: Dynamic Combination of Detector Scores for Outlier Ensembles
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (***KDD Workshop** on Outlier Detection De-constructed*), 2018, London, UK.
Extended version published in *SDM 2019*, renamed to LSCP.

3. Yue Zhao, Maciej K. Hryniewicki
XGBOD: Improving Supervised Outlier Detection with Unsupervised Representation Learning
IEEE International Joint Conference on Neural Networks (***IJCNN***), 2018, Rio, Brazil.

2. Yue Zhao, Maciej K. Hryniewicki, Francesca Cheng, Boyang Fu, Xiaoyu Zhu
Employee Turnover Prediction with Machine Learning: A Reliable Approach
Intelligent System Conference (***Intellisys***), 2018, London, UK.

1. Yue Zhao†, Zhongtian Qiu†, Yiqing Yang†, Weiwei Li†, Mingming Fan
An Empirical Study of Touch-based Authentication Methods on Smartwatches
ACM International Symposium on Wearable Computers (***ISWC***), 2017, Maui, USA.

Internship Experience

NortonLifeLock Research Group
Machine Learning Research Intern — 2022

Microsoft Research
Machine Learning Research Intern — 2022

Stanford University, Computer Science Department
Visiting Student Researcher (Prof. Jure Leskovec) — 2021

IQVIA, Analytics Center of Excellence
Machine Learning Research Intern — 2020

Siemens PLM Software USA
Software Engineer (Intern & Contract) — Mar. 2012 - Dec. 2014

Teaching Experience

University of Southern California — Los Angeles, CA
Instructor — Fall 2026 (scheduled)
CSCI 566 Deep Learning and Its Applications

Instructor Spring 2026
CSCI 699 Adversarial and Trustworthy Foundation Models

Instructor Spring 2025
CSCI 566 Deep Learning and Its Applications

Instructor Spring 2024
CSCI 566 Deep Learning and Its Applications

Carnegie Mellon University Pittsburgh, PA

Teaching Assistant Fall 2022
Managing Digital Business (Prof. David Riel)

Teaching Assistant & co-Instructor (lectures on AutoML and MLSys) Spring 2022 – Fall 2020
Intro to Artificial Intelligence (Prof. David Steier)

Teaching Assistant Spring 2022
Digital Transformation (Prof. David Riel)

Teaching Assistant (helping on course topics) Fall 2021
Statistics for IT Managers (Prof. Daniel Nagin)

University of Toronto Toronto, ON

Teaching Assistant & Lab Session Instructor Fall 2015
Embedded Systems (Prof. Philip Anderson)

University of Cincinnati Cincinnati, OH

Teaching Assistant & Lab Session Instructor Fall 2014
Intro to Programming (Prof. George Purdy)

Ph.D. Students

- Haoyan Xu (USC, ECE Ph.D. Candidate, 2024 Spring-), co-advised by Mengyuan Li, Capital One Fellowship
- Yuehan Qin (USC, CS Ph.D. Candidate, 2024 Fall-)
- Tiankai Yang (USC, CS Ph.D., 2024 Fall-)
- Shawn Li (USC, CS Ph.D. Candidate, 2024 Fall-), Capital One Fellowship, Amazon ML Fellowship
- Jiate Li (USC, CS Ph.D., 2025 Fall-)
- Wei Yang (USC, CS Ph.D. Candidate, 2026 Fall-), co-advised by Jesse Thomason and Xuezhe Ma
- Yi Nian (USC, CS Ph.D., incoming)
- Chenxiao Yu (USC, CS Ph.D., incoming)
- Zixiang Xu (USC, CS Ph.D., incoming)

Student Committee

Term	Student	Committee
Fall 2026	Chen Chu (USC, CS Ph.D.)	Qualification
Summer 2026	Defu Cao (USC, CS Ph.D.)	Defense
Spring 2026	Longchao Da (ASU, CS Ph.D.)	Defense
	Yahan Li (USC, CS Ph.D.)	Qualification
	Jiageng Mao (USC, CS Ph.D.)	Thesis
	Xinyue Cui (USC, CS Ph.D.)	Qualification
	Jiawei Yang (USC, CS Ph.D.)	Qualification + Thesis
	Elnaz Rahmati (USC, CS Ph.D.)	Qualification
Fall 2025	Hao Dong (ETH Zurich, CS Ph.D.)	Defense
	Haonan Wang (USC, ECE Ph.D.)	Defense
Spring 2025	Mengxi Wu (USC, CS Ph.D.)	Qualification
	Xingrui Wang (USC, ME Ph.D.)	Defense
Fall 2024	Alex Bisberg (USC, CS Ph.D.)	Thesis
Summer 2024	Gengyu Rao (USC, CS Ph.D.)	Defense
	Mehrdad Kiamari (USC, ECE Ph.D.)	Defense
Spring 2024	Maria Despoina Siampou (USC, CS Ph.D.)	Qualification
	Yuan Meng (USC, ECE Ph.D.)	Defense
	Hassan Hamad (USC, ECE Ph.D.)	Qualification
	Yizhou Zhang (USC, CS Ph.D.)	Thesis
	Haoming Li (USC, CS Ph.D.)	Qualification
	Arash Hajisafi (USC, CS Ph.D.)	Qualification
Fall 2023	Yi Chien Lin (USC, ECE Ph.D.)	Qualification
	Yuke Zhang (USC, ECE Ph.D.)	Qualification

Services

Community Outreach

- Founder & Maintainer, CSPhD.org: a non-profit community for CS/AI/EE/Stats PhDs (since 2022, several thousand members, 500+ PhD/RA/postdoc openings), supporting students from before enrollment to the faculty job market.

Conference/Workshop Organizing Roles

- Doctoral Consortium Co-Chair, AAAI 2027
- Future Faculty Highlights Co-Chair, IEEE BigData 2026
- Publicity Co-Chair, WSDM 2027
- Workflow Co-Chair, KDD 2023
- Co-organizer, SURGeLLM: Structured Understanding, Retrieval, and Generation in LLMs era Workshop @ ACL 2026
- Co-organizer, AgentSkills '26: The First Workshop on Agent Skills @ CAIS Conference 2026
- Co-organizer, AI for Financial Fraud Detection & Prevention Workshop @ 6th ACM International Conference on AI in Finance

Funding Proposal Review Roles

- National Science Foundation (NSF)
- Dutch Research Council (NWO)

Journal Editorial Roles

- Action Editor, Transactions on Machine Learning Research (TMLR), 2026–present
- Associate Editor, ACM Transactions on AI for Science (TAIS), 2025–present
- Associate Editor, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024–present
- Action Editor, Journal of Data-centric Machine Learning Research (DMLR), 2024–present

Conference/Workshop AC and Reviewer Roles

- ICLR 2025 (AC), ICLR 2026 (AC)
- AAAI 2021, 2022, 2023, 2025 (Senior PC), 2026 (Senior PC)
- ICML 2024, 2025 (AC), 2026 (AC)
- NeurIPS 2021, 2022, 2023, 2025 (AC), 2026 (AC)
- AISTATS 2024, 2025 (AC)
- MLSys 2024, 2026
- KDD 2020, 2021, 2022, 2023, 2026
- ACL Rolling Review (ARR) 2026
- IJCAI 2022, 2023
- AAAI Demonstrations 2021, 2022
- MICCAI 2020, 2021, 2022
- ICDM 2020
- KDD Workshop on Outlier Detection and Description (ODD), 2021
- KDD Workshop on Anomaly and Novelty Detection (ANDEA), 2021, 2022
- IJCAI Workshop on Artificial Intelligence for Anomalies and Novelties (AI4AN), 2020, 2021
- INFORMS Workshop on Data Science 2021

Journal Review Roles

- Journal of Machine Learning Research (JMLR)
- PNAS Nexus
- Machine Learning
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Intelligent Systems
- IEEE Journal on Selected Areas in Communications (J-SAC)
- Data Mining and Knowledge Discovery (DMAI)
- ACM Transactions on Management Information Systems (TMIS)
- Knowledge and Information Systems (KAIS)
- INFORMS Journal on Computing (IJOC)
- Big Data
- Artificial Intelligence Review (AIRE)
- Neurocomputing
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)
- IEEE Network Magazine
- IEEE Computational Intelligence Magazine (CIM)
- BioData Mining
- European Journal of Management and Business Economics (EJMBE)
- The Journal of Open Source Software (JOSS)

Talks and Lectures

NVIDIA Agent & Security	*Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker*	May. 2026
Amazon AI Security	*Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker*	May. 2026
USC symposium on Frontiers of ML/AI	*Towards Robust AI: Advances in Outlier and OOD Detection*	Mar. 2025
NUS Tea Talk	*Towards Robust AI: Advances in Outlier and OOD Detection*	Jan. 2025
SFU@NeurIPS'24	*Towards Robust AI: Advances in Outlier and OOD Detection*	Dec. 2024
KAIST	*Unsupervised Model Selection: Automation with Meta-learning and LLMs*	Nov. 2024
Kennesaw State University	*Unsupervised Model Selection: Automation with Meta-learning and LLMs*	Oct. 2024
LinkedIn Anti-Abuse AI	*Outlier Detection: Automation, Systems, and GenAI*	Aug. 2024
Amazon Security AI	*Outlier Detection: Automation, Systems, and GenAI*	Aug. 2024
New York University	*Outlier Detection: Automation, Systems, and GenAI*	Aug. 2024
University of Washington	*Outlier Detection: Automation, Systems, and GenAI*	Jun. 2024
Microsoft	*Outlier Detection: Automation, Systems, and GenAI*	Jun. 2024
USC Retreat on AI and Engineering Safety	*Safety Measures for LLMs*	Apr. 2024
Visa Research	*Towards Reproducible, Automated, and Scalable AD*	Apr. 2024
USC Symposium on Frontiers of Generative AI	*Generative AI for Anomaly Detection*	Mar. 2024
AAAI New Faculty Highlights (invited)	*Towards Reproducible, Automated, and Scalable AD*	Feb. 2024
U of Nevada, Las Vegas	*Automated and Scalable ML Algorithms and Systems*	Oct. 2023
Samsung Seminar	*Automated and Scalable Anomaly Detection Systems*	Aug. 2023
KDD SoCal Day	*Enable Applications by ML with Noisy Inputs*	Aug. 2023
CMU Catalyst	*How (Not) to Fail Your Academic Job Search*	May. 2023
KAUST	*Automated and Scalable ML Algorithms and Systems*	Apr. 2023
Emory University	*Automated and Scalable ML Algorithms and Systems*	Apr. 2023
USC	*Automated and Scalable ML Algorithms and Systems*	Mar. 2023
UC Davis	*Automated and Scalable ML Algorithms and Systems*	Mar. 2023
Stony Brook University	*Automated and Scalable ML Algorithms and Systems*	Feb. 2023
University of Chicago	*Automated and Scalable ML Algorithms and Systems*	Feb. 2023
UC Merced	*Automated and Scalable ML Algorithms and Systems*	Feb. 2023
CMU PDL Meeting	*Automated and Scalable ML Algorithms and Systems*	Jan. 2023
CMU Data Science Seminar	**Guest Lecture** *Automated Anomaly Detection*	Nov. 2022
LoG Seminar	*Large-scale Graph Anomaly Detection*	Oct. 2022
Intuit	*Anomaly Detection for Financial Risk Modeling*	Aug. 2022
Rice University	*Large-scale Anomaly Detection with Automation*	Sep. 2022
Microsoft Research	*Weakly-supervised Anomaly Detection*	Sep. 2022
Wells Fargo	*Anomaly Detection for Financial Risk Modeling*	Aug. 2022
Columbia University	**Guest Lecture** *Anomaly Detection*	Jul. 2022
Morgan Stanley	*Automated Outlier Detection*	Jun. 2022
Microsoft Research	*Automated Outlier Detection*	Jun. 2022
Morgan Stanley	*Large-scale Anomaly Detection Systems*	Mar. 2022
Rutgers Business School	*Outlier Model Selection*	Mar. 2022
Tesla	*Large-scale Anomaly Detection Systems*	Feb. 2022
Catalyst, CMU	*Systems for Data Mining Algorithms*	Dec. 2021
E&Y Canada	*ML applications in Data Analytics*	Oct. 2021
University of Nottingham	*General Machine Learning Applications*	Jan. 2021